

MUHAIMINUL ASHRAF

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Full-stack engineer with a hardware foundation — experienced building AI-integrated web applications, backend pipelines, and the systems underneath them. Co-op experience spans R&D software tooling, e-commerce platform development, and medical device automation.

SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, SQL, Bash **DevOps / Tooling:** Git, Docker, CI/CD (GitHub Actions), Linux, AWS, Jira
Web / Backend: Next.js, React, Node.js, Flask, REST APIs, Prisma ORM, Vercel AI SDK **Databases:** PostgreSQL, MySQL, Supabase
Hardware: Embedded Linux, ESP32/STM32, PCB Design (KiCad), RF Systems, FPGA

EXPERIENCE

Grovara <i>Software Engineer (Contract)</i>	Jan 2025 – Jun 2025 Remote
- Rebuilt a B2B wholesale e-commerce platform from legacy CakePHP to a modern React and TypeScript stack, owning full-stack delivery from responsive UI components and client-side state management through API integration and deployment. - Designed and deployed RESTful API endpoints in Node.js backed by PostgreSQL to automate inventory data synchronization across the platform, eliminating manual reconciliation and supporting thousands of daily transactions with consistent data integrity.	
Kostas Research Institute <i>Engineering R&D Co-Op</i>	Jan 2025 – Aug 2025 Burlington, MA
- Wrote Bash scripts to automate bring-up and configuration of Linux SBCs across a multi-board RF antenna test environment, standardizing signal capture, log collection, and hardware fault detection, cutting per-board setup time from hours to minutes. - Built MATLAB analysis tooling to process raw spectrum analyzer captures from experimental antenna field tests, implementing filtering and demodulation pipelines to decode on-board signals and classify novel waveform data across multiple field campaigns. - Presented technical findings as documentation and live demos for non-technical stakeholders, supporting \$7M+ in grant renewals.	
Insulet <i>Systems Engineer – QA and Design Verification Co-Op</i>	Jan 2024 – Aug 2024 Acton, MA
- Built a Python OCR automation script for mobile onboarding verification, increasing the team's daily testing throughput by 40%. - Parsed device telemetry logs from AWS S3 using Python to identify fault signatures and support root-cause analysis. - Ran end-to-end design verification for the Omnipod insulin delivery device, documenting and escalating three critical reliability issues before production release and contributing to CE marking compliance for the European market launch.	

LEADERSHIP

TheBoard — Head of Design	Jan 2024 – Jun 2024
Managed a 5-person design team for a student-founded edtech startup, running weekly design reviews and feature brainstorming sessions while owning UX/UI direction from wireframes through prototype.	

PROJECTS

poKeDB — Full-Stack Web Application pokedb-one.vercel.app

- Iterated through three data-fetching architectures — client-side API calls, PostgreSQL with Redis caching, and build-time static generation, reducing page load from ~3s to instant across 1,000+ routes.
- Designed a normalized PostgreSQL schema with Prisma ORM for many-to-many Pokémon relationships; seeded via ETL from the PokéAPI, migrated to Supabase, and deployed on Vercel with GitHub Actions CI/CD.
- Built 15+ reusable React components (i.e. sprite carousels, game-version-tabbed Pokédex selectors, type effectiveness charts) with dynamic theming that adapts per generation.

Cortex — AI-Powered Adaptive Learning Platform

- Built an adaptive learning platform that generates personalized tutoring sessions via the Anthropic API: backed by Supabase (PostgreSQL + Auth), Prisma ORM, and a Next.js/TypeScript frontend with streaming AI responses deployed on Vercel.
- Designed a deterministic session orchestrator that selects pedagogical modes, dynamically constructs LLM prompts from learner state, and parses structured JSON responses with Zod validation — keeping the UI predictable and the learning flow testable.
- Implemented a per-user knowledge graph with SM-2 spaced repetition, mastery-based difficulty progression, and a d3.js force-directed topic map that surfaces weak areas and adapts review scheduling to each learner's retention patterns in real time.

AERIS — Full-Stack IoT Platform

- Built an IoT environmental monitoring platform end-to-end: ESP32 C++ firmware collecting telemetry over I2C/UART with SRAM buffering for offline resilience, a Flask REST API ingesting sensor data into MySQL for time-series queries, and a dashboard surfacing trend visualizations and threshold-based safety alerts.

EDUCATION

Northeastern University <i>B.Sc Computer Engineering</i>	Sep 2021 – Dec 2025 Boston, MA
Horace Mann School <i>High School Diploma</i>	Bronx, NY